Course Announcements

Due Dates

- **Tonight** (3/12; 11;59 PM): A4 (late deadline still Friday)
- next Wednesday (3/19):
  - Final project*
  - video*
  - team eval survey (will be available Fri) - ideally complete *after* project is completed
  - post-course survey (will be available Fri)

Notes:

- Please complete your SETs
- Labs this week are happening...but there's nothing to turn in; go get project questions answered!
- No office hours during finals week

How to Be Wrong

Shannon E. Ellis, Ph.D
UC San Diego

Department of Cognitive Science
sellis@ucsd.edu

Lecture concept and slides built off of Jason Fleischer's lecture of the same name

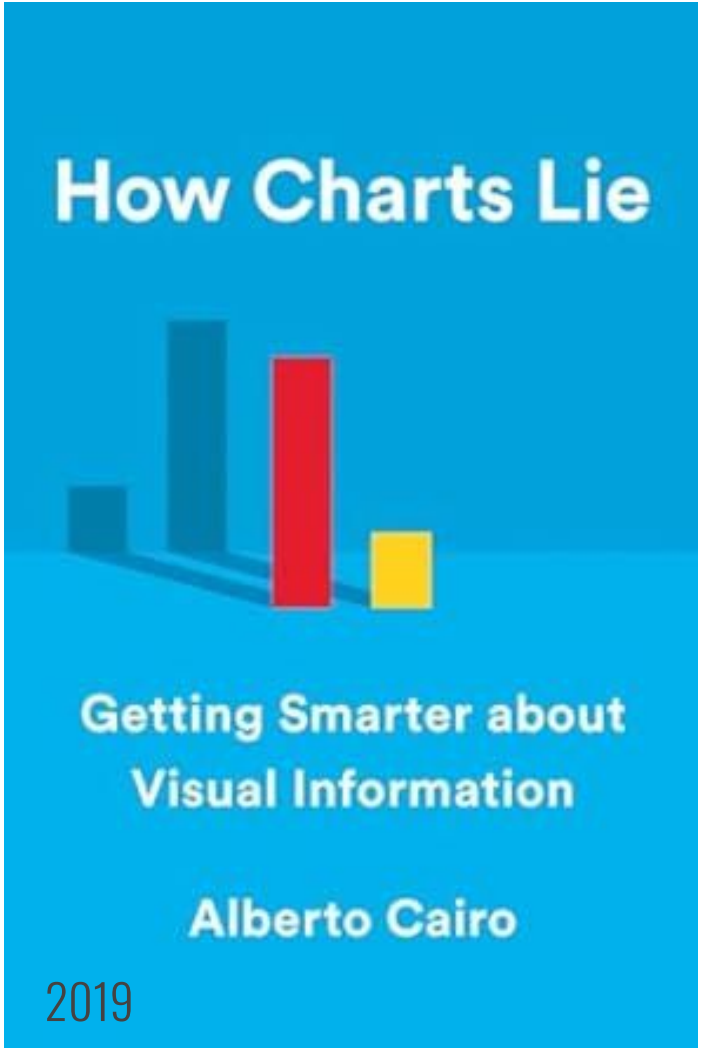
Importance of communication (previously)

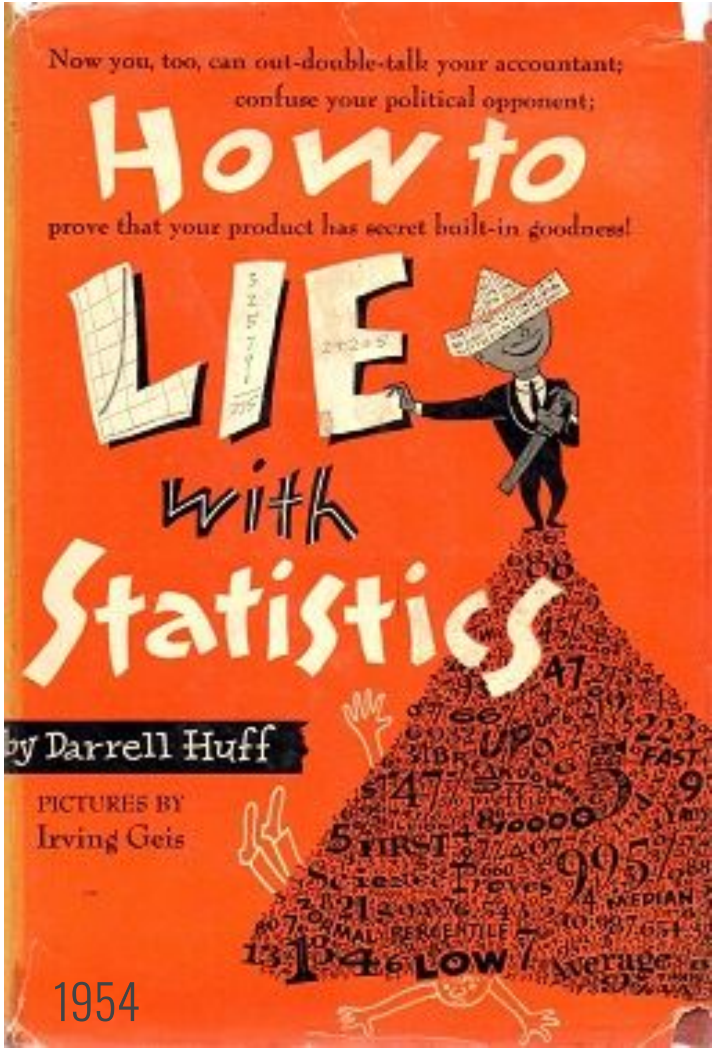
Errors of human cognition

Errors of measurement

Errors of broken tools

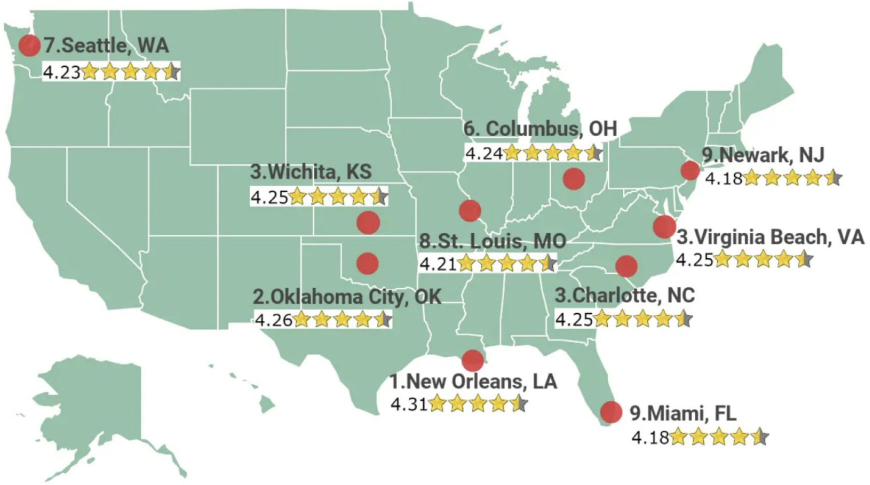
Errors of analysis


How Charts Lie
Getting Smarter about Visual Information
Alberto Cairo
2019


Now you, too, can out-double-talk your accountant;
confuse your political opponent;
How to
prove that your product has secret built-in goodness!
LIE
with
Statistics
by Darrell Huff
PICTURES BY
Irving Geis
1954

Top Cities For BBQ in the U.S.

An analysis of TripAdvisor restaurant reviews by chefspencil.com



Worst Cities For BBQ in the U.S.

An analysis of TripAdvisor restaurant reviews by chefspencil.com

https://www.wspa.com/news/tripadvisor-sets-the-record-straight-after-bbq-ranking-left-kansas-city-out-of-top-30/

Errors of human cognition

1A (left side of room)

https://forms.gle/oLgNaWVeC5QnC7YQ8

1B (right side of room)

https://forms.gle/rnKjCKKBz5avEGpi6

Anchoring bias

The tendency for decision makers to start their decision process from a specific point of reference and subsequently make insufficient adjustments from the anchored point.

Possible technique to reduce the bias:

- Document the rationale for a decision
- Learn to anticipate and correct the bias
- Develop expectations through experience

https://forms.gle/TY5VRNr4Ng8MTj2G6

Exercise 2

Overconfidence Bias

- Tendency for decision makers to overestimate their own abilities
- Generally two types: overplacement & illusion of explanatory depth
- Some examples (from wikipedia): Chernobyl, Titanic, Challenger Disaster, Horizon Oil Spill, Titan Submersible impostion

A list of names for Exercise 3

Travis Kelce

Olivia Rodrigo

Blake Lively

Rock Hudson

Selena Gomez

Kesha

Taylor Swift

Charles Barkley

Cardi B

Bruno Mars

David Seaman

Richard Grieco

Sinbad

Steve Winwood

Megan Thee Stallion

Barry White

Steven Tyler

Beyonce

Millie Bobby Brown

Lady Gaga

Robert Duvall

https://forms.gle/8CxQb7aP4V4mK6r96

Exercise 3

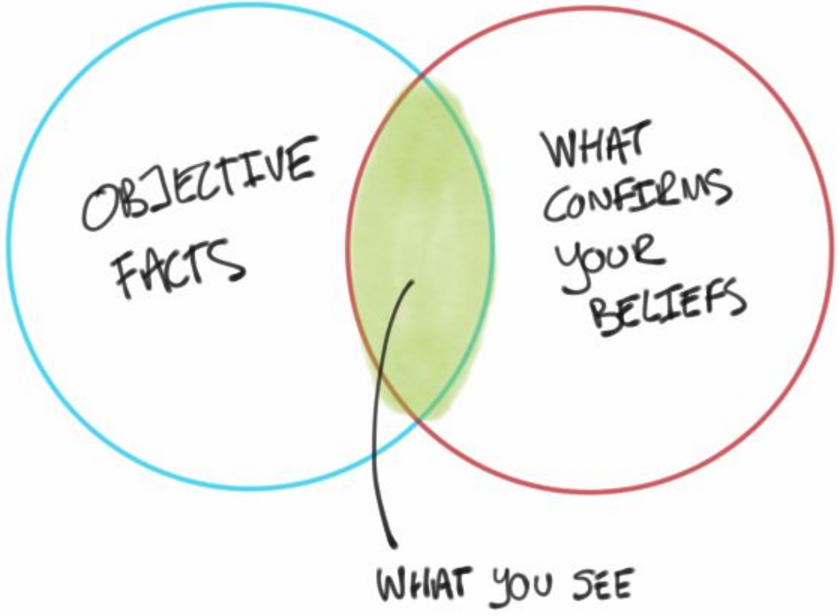
Availability bias

The tendency for decision makers to consider information that is easily retrievable from memory as being more likely, more relevant, and more important for judgment

What we saw:

- Recent celebrity names are easier to remember/recognize
- More generally: memory is weird and wonky!


CHAINSAWSUIT.COM
i've heard the rhetoric from both sides... time to do my own research on the real truth
Googie
hotly debated topic
Found 80,000 results.
Literally the first link that agrees with what you already believe
Completely supports your viewpoint without challenging it in any way
Another link
...jackpot


OBJECTIVE FACTS
WHAT CONFIRMS YOUR BELIEFS
WHAT YOU SEE

Cognitive Biases discussed so far

Anchoring bias: pre-analysis ideas influencing post-analysis answer

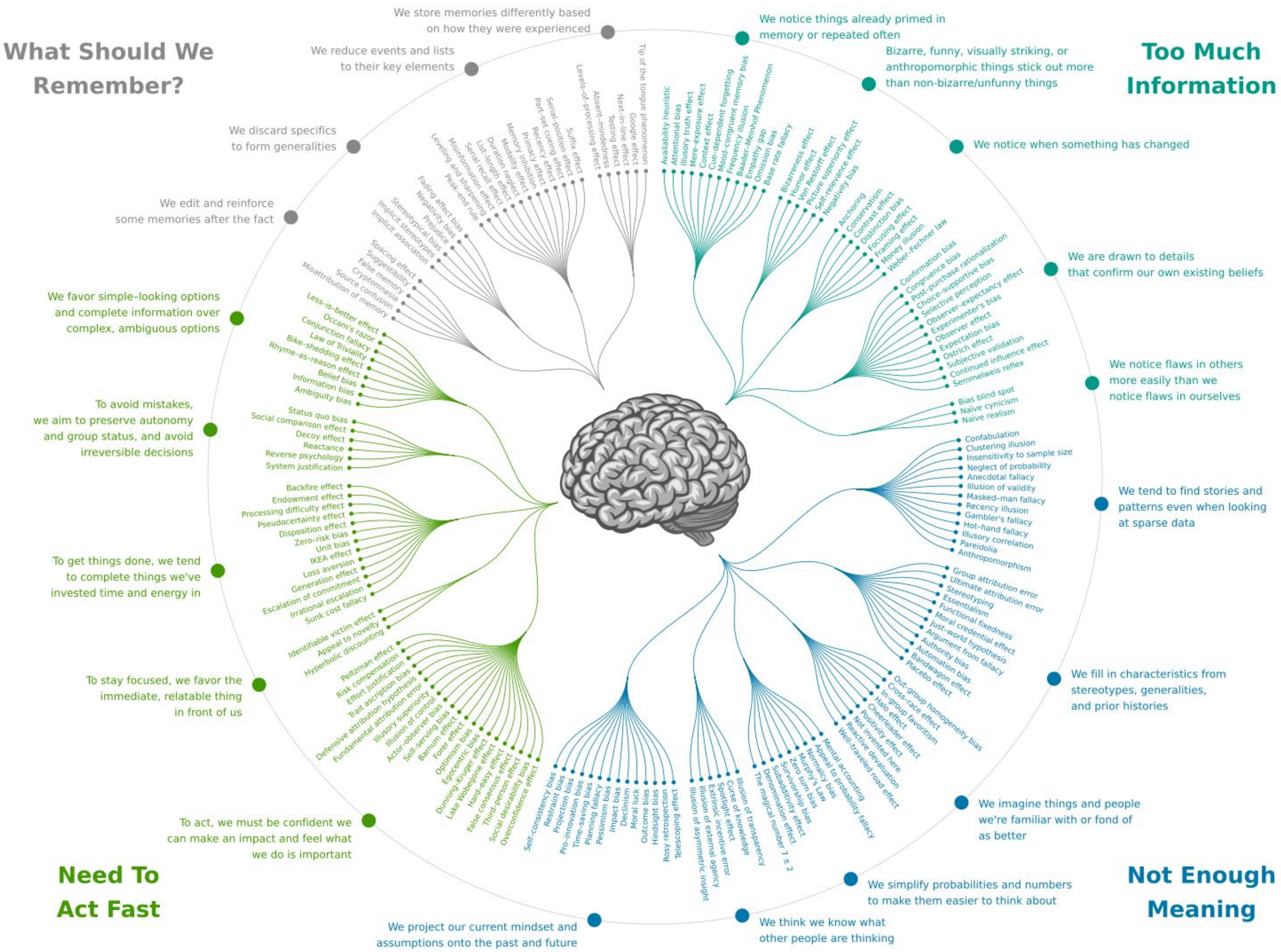
Illusion of explanatory depth: thinking you understand when you're just smiling and nodding (or guessing without information)

Availability bias: familiar terms are remembered better than novel ones

Confirmation bias: more quickly agreeing with information that agrees with what you already believe/know

THE COGNITIVE BIAS CODEX

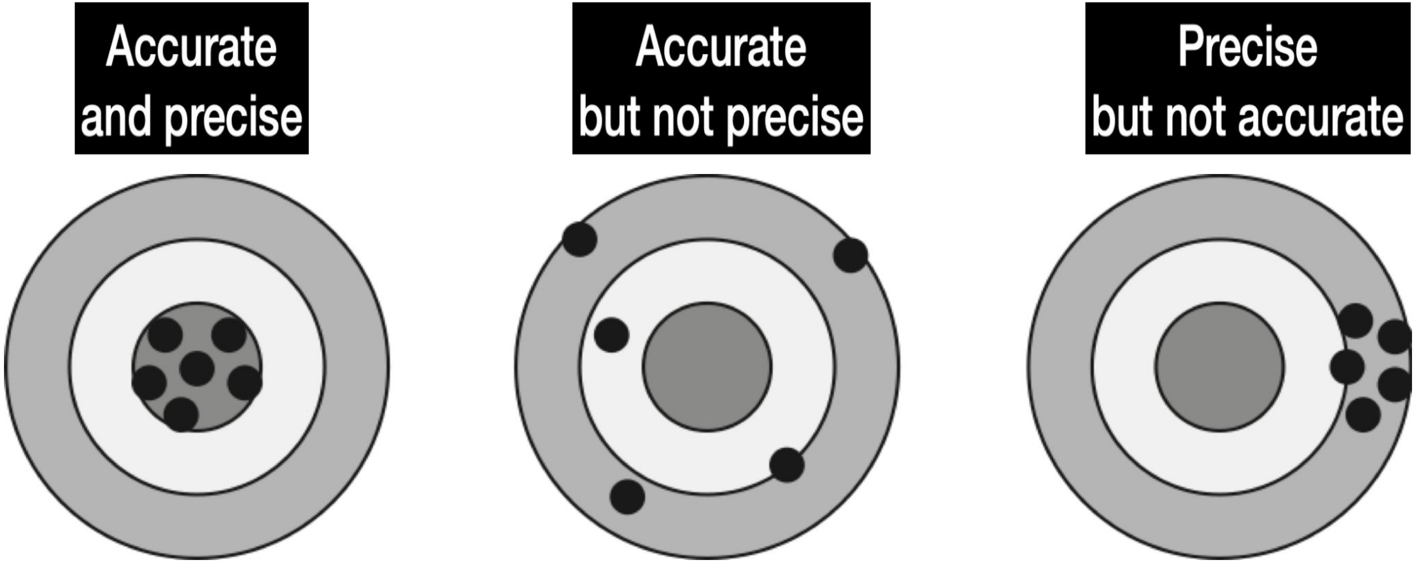
https://upload.wikimedia.org/wikipedia/commons/6/65/Cognitive_bias_codex_en.svg



Fighting Biases

- Know they exist
- Instruction, feedback, and practice
- Know procedures to test your hypotheses
- Question yourself and your ideas aggressively
- Have procedures that for you to question basis of decisions
- Find others willing to do the same (beware groupthink …hello, confirmation bias)

Errors of measurement


Accurate and precise
Accurate but not precise
Precise but not accurate

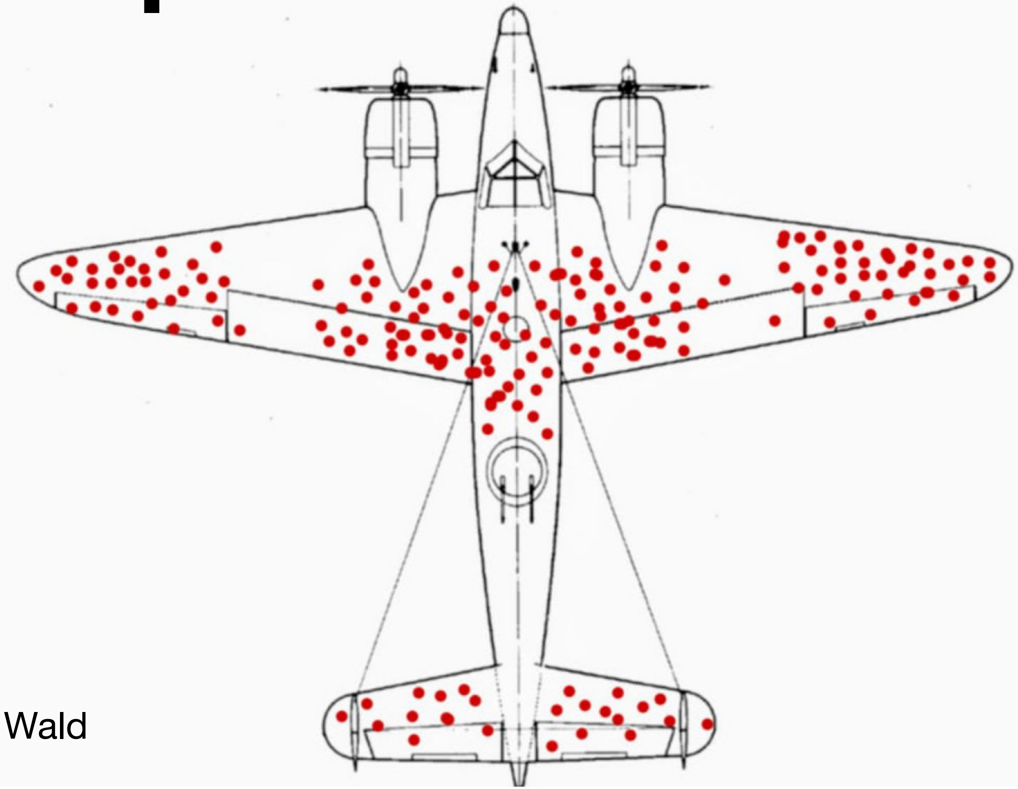
Proxy Measurements

For when we can’t measure directly what we care about, but we can measure something related

- Unemployment rate → our general economy
- Gross domestic product → standard of living
- BMI → health
- Your suggestions?
- Grades → learning

Do we understand what we are measuring?

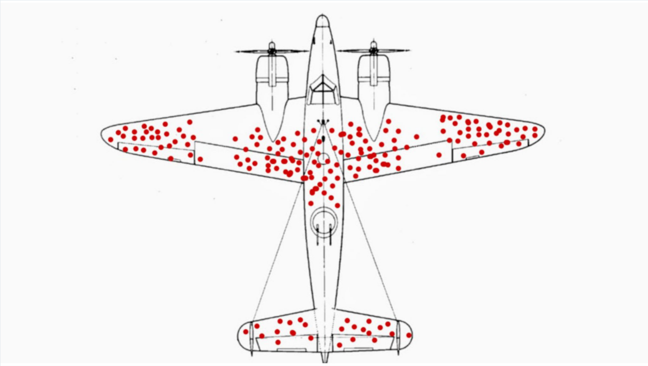
Survivorship bias



Abraham Wald

Survivorship Bias

https://forms.gle/CQm8smTuSCZbhezS7



Where would you put the armor?

A
Wings

B
Tail

C
Front of plane

D
Middle of wings

E
Middle of tail

Are we measuring what's relevant?

A This reflects my experience

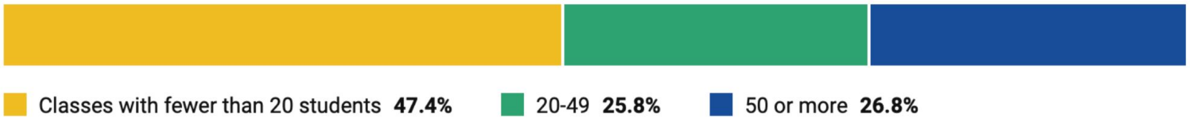
B This kind of reflects my experience

C This does not reflect my experience

Academic Life at University of California--San Diego

The student-faculty ratio at University of California--San Diego is 19:1, and the school has 47.4% of its classes with fewer than 20 students. The most popular majors at University of California--San Diego include: Biology, General; Mathematics; Economics; International/Global Studies; and Computer Science. The average freshman retention rate, an indicator of student satisfaction, is 94%.

Class Sizes

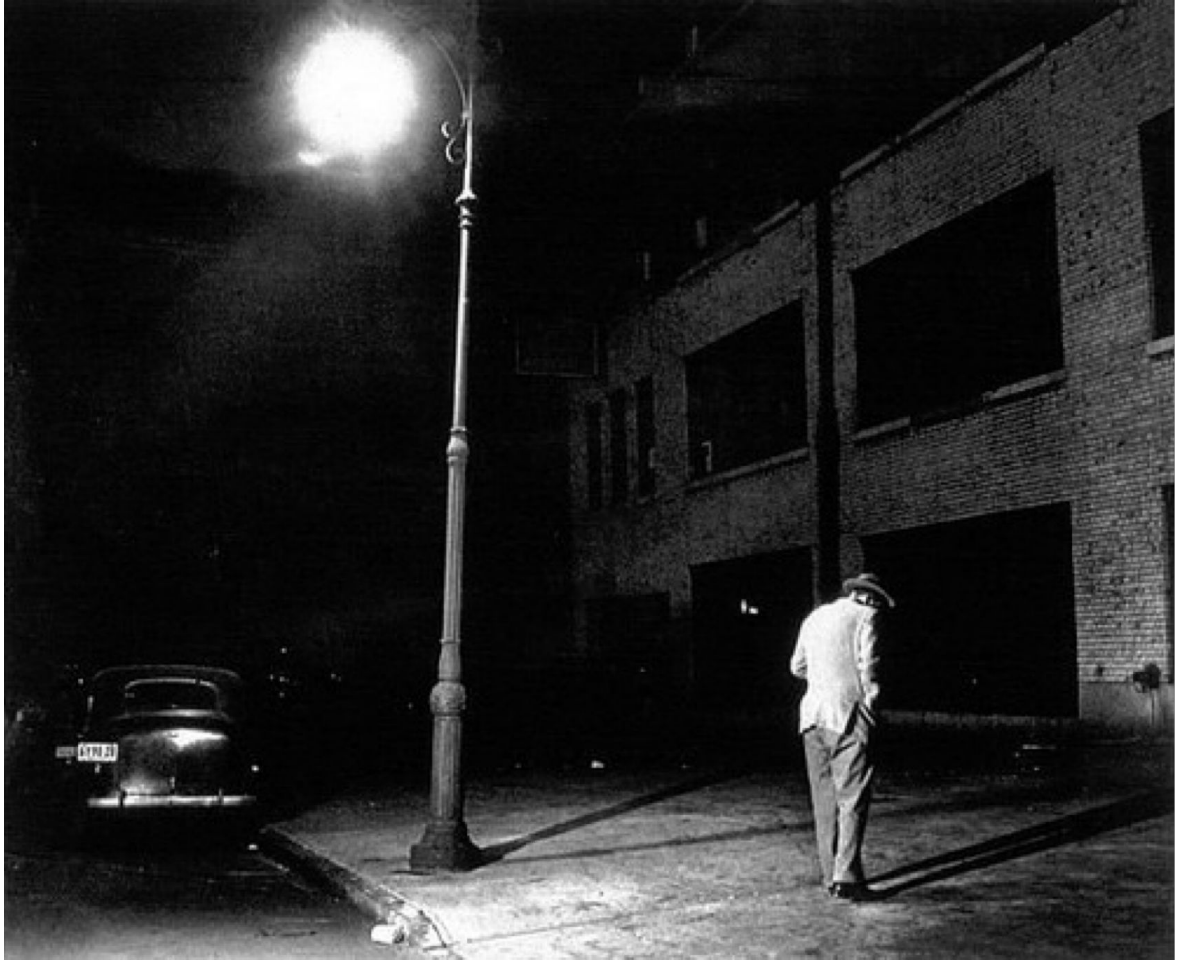


Student-faculty ratio	**19:1**
4-year graduation rate	**65%**

		% of classes with this many students	Cumulative %	Fraction of classes with this many students * min number of students in that class type	% of students in these classes (normalized version of column to the left)	
	2-9 students:	12%	12%	0.24	0.67%	
	10-19 students	32%	44%	3.2	8.95%	
Median class size as experienced by faculty	20-29 students:	14%	58%	2.8	7.83%	
	30-39 students:	8%	66%	2.4	6.72%	
	40-49 students:	4%	70%	1.6	4.48%	
	50-99 students:	11%	81%	5.5	15.39%	
Median class size as experienced by students	Over 100 students:	20%	101%	20	55.96%	
			Sum:	35.74		
	Data from https://www.collegedata.com/college/University-of-California-San-Diego/					

The Lamppost Problem

aka. Streetlight Effect
aka. The Drunkard's Search
aka. Picking a crap proxy



Goodhart's law

Once a metric becomes a goal, you're hosed

Volkswagen emissions scandal

文A 26 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

"Dieselgate" and "Emissionsgate" redirect here. For other diesel emissions scandals, see Diesel emissions scandal.

The **Volkswagen emissions scandal**, sometimes known as **Dieselgate**[23][24] or **Emissionsgate**,[25][24] began in September 2015, when the United States Environmental Protection Agency (EPA) issued a notice of violation of the Clean Air Act to German automaker Volkswagen Group.[26] The agency had found that Volkswagen had intentionally programmed turbocharged direct injection (TDI) diesel engines to activate their emissions controls only during laboratory emissions testing, which caused the vehicles' NO_x output to meet US standards during regulatory testing. However, the vehicles emitted up to 40 times more NO_x in real-world driving.[27] Volkswagen deployed this software in about 11 million cars worldwide, including 500,000 in the United States, in model years 2009 through 2015.[28][29][30][31]

Volkswagen emissions scandal

A 2010 Volkswagen Golf TDI displaying "Clean Diesel" at the Detroit Auto Show

Search engine optimization

文A 65 languages

Article Talk Read View source View history Tools

From Wikipedia, the free encyclopedia

"SEO" redirects here. For other uses, see Seo.

Search engine optimization (**SEO**) is the process of improving the quality and quantity of website traffic to a website or a web page from search engines.[1][2] SEO targets unpaid traffic (known as "natural" or "organic" results) rather than direct traffic or paid traffic. Unpaid traffic may originate from different kinds of searches, including image search, video search, academic search,[3] news search, and industry-specific vertical search engines.

As an Internet marketing strategy, SEO considers how search engines work, the computer-programmed algorithms that dictate search engine behavior, what people search for, the actual search terms or keywords typed into search engines, and which search engines are preferred by their targeted audience. SEO is performed because a website will receive more visitors from a search engine when websites rank higher on the search engine results page (SERP). These visitors can then potentially be converted into customers.[4]

Part of a series on
Internet marketing
Search engine optimization
Local search engine optimisation
Social media marketing
Email marketing
Referral marketing
Content marketing
Native advertising
Search engine marketing
Pay-per-click
Cost per impression
Search analytics
Web analytics
Display advertising
Ad blocking
Contextual advertising

Errors of assumptions around tools

Excel

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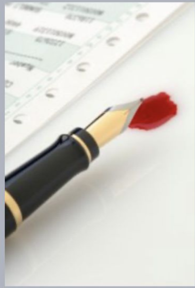
Conference Abstracts, Papers & Indexes
Conference Reports & Videos
Delegates
Constitution
History
Our Talks & Presentations
Quotable Quotes
Press & Website
Committees
Discussion group
Training Videos
Humour
Sponsors
Useful Links
Contact

EuSpRIG
European Spreadsheet Risks Interest Group

EuSpRIG Horror Stories

Spreadsheet mistakes - news stories



Public reports of spreadsheet errors have been sought out on behalf of EuSpRIG by Patrick O'Beirne of Systems Modelling for many years. There are very many reports of spreadsheet related errors and they seem to appear in the global media at a fairly consistent rate.

These stories illustrate common problems that occur with the uncontrolled use of spreadsheets. In many cases, we identify the area of risk involved and then say how we think the problem might have been avoided.

Stories are identified by those who kindly collated and sorted them:

POB: Patrick O'Beirne, Eusprig chair

FH: Felienne Hermans (winner of the 2011 David Chadwick student prize and now an assistant professor at Delft University of Technology).

NS: Tie Cheng, a EuSpRIG committee member.

MPC: Mary Pat Campbell, an actuary, trainer, and a member of the EuSpRIG Discussion group.

Identifier:	POB2001
Title:	Data not controlled, 16000 UK Covid-19 test results lost for a week
Source:	https://www.bbc.co.uk/news/technology-54423988
Release Date:	08 October 2020
Risk:	Lives put at risk because the contact-tracing process had been delayed
Discrepancy:	16,000 test cases in a week

Excel: Why using Microsoft's tool caused Covid-19 results to be lost
"The badly thought-out use of Microsoft's Excel software was the reason nearly 16,000 coronavirus cases went unreported in England. [The labs] filed their [result logs] results in the form of text-based lists - known as CSV files - without issue. PHE had set up an automatic process to pull this data together into Excel templates so that it could then be uploaded to a central system. The problem is that [Public Health England] PHE's own developers picked an old file format to do this - known as XLS. As a consequence, each template could handle only about 65,000 rows of data rather than the one million-plus rows that Excel is actually capable of. And since each test result created several rows of data, in practice it meant that each template was limited to about 1,400 cases. When that total was reached, further cases were simply left off. To handle the problem, PHE is now

http://www.eusprig.org/horror-stories.htm

MICROSOFT \ REPORT \ SCIENCE

99

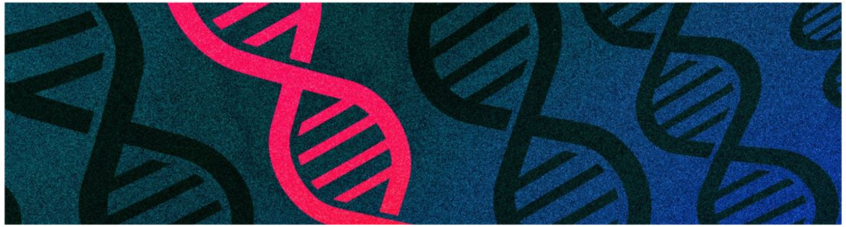
Scientists rename human genes to stop Microsoft Excel from misreading them as dates

Sometimes it's easier to rewrite genetics than update Excel

By James Vincent | Aug 6, 2020, 8:44am EDT

Listen to this article

SHARE



https://www.theverge.com/2020/8/6/21355674/human-genes-rename-microsoft-excel-misreading-dates

What about when Excel is used as intended?

15k spreadsheets
97M cells
20M formulas

Enron's Spreadsheets and Related Emails: A Dataset and Analysis

Felienne Hermans
Delft University of Technology
Mekelweg 4
2628 CD Delft, the Netherlands
f.f.j.hermans@tudelft.nl

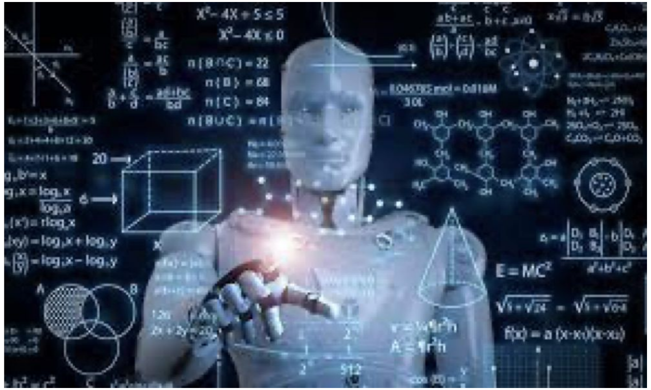
Emerson Murphy-Hill
North Carolina State University
890 Oval Drive
Raleigh, North Carolina, USA
emerson@csc.ncsu.edu

TABLE III
SPREADSHEETS CONTAINING EXCEL ERRORS IN THE ENRON SET

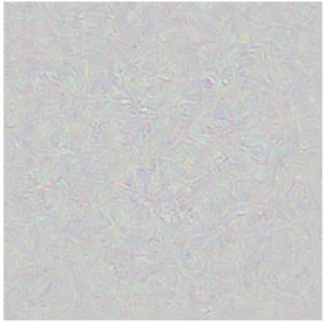
Error type	Spreadsheets	Formulas	Unique Ones
#DIV/0!	580	76,656	4,779
#N/A	635	948,194	6,842
#NAME?	297	33,9365	29,422
#NUM!	52	4,087	178
#REF!	931	18,3014	6824
#VALUE!	423	11,1024	1751
Total	2,205	1,662,340	49,796

24% of spreadsheets with formulas had errors

Excel



Learning the irrelevant
ML loves the shortcut



Article: Super Bowl 50

Paragraph: *"Peython Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver's Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had a jersey number 37 in Champ Bowl XXXIV."*

Question: *"What is the name of the quarterback who was 38 in Super Bowl XXXIII?"*

Original Prediction: John Elway

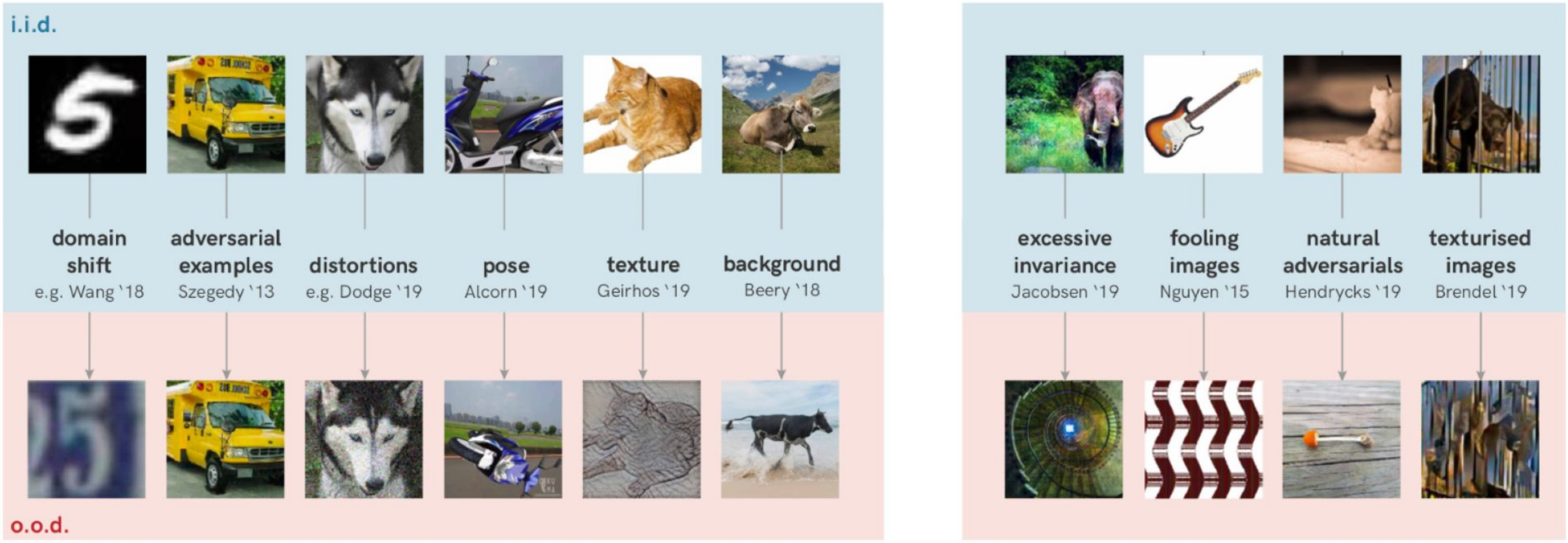
Prediction under adversary: Jeff Dean

Task for DNN	Caption image	Recognise object	Recognise pneumonia	Answer question
Problem	Describes green hillside as grazing sheep	Hallucinates teapot if certain patterns are present	Fails on scans from new hospitals	Changes answer if irrelevant information is added
Shortcut	Uses background to recognise primary object	Uses features irrecognisable to humans	Looks at hospital token, not lung	Only looks at last sentence and ignores context

https://thegradient.pub/shortcuts-neural-networks-love-to-cheat/

same category for humans
but not for DNNs (intended generalisation)

same category for DNNs
but not for humans (unintended generalisation)



Tell me a novel joke about the singer Madonna.

Why did Madonna study geometry?

Because she wanted to learn how to strike a pose in every angle!

10 Ways GPT-4 Is Impressive but Still Flawed

OpenAI has upgraded the technology that powers its online chatbot in notable ways. It's more accurate, but it still makes things up.

By Cade Metz and Keith Collins

Cade Metz asked experts to use GPT-4, and Keith Collins visualized the answers that the artificial intelligence generated.

March 14, 2023

Errors of analysis

Preregistration and publishing negative results

"Data available upon reasonable request"

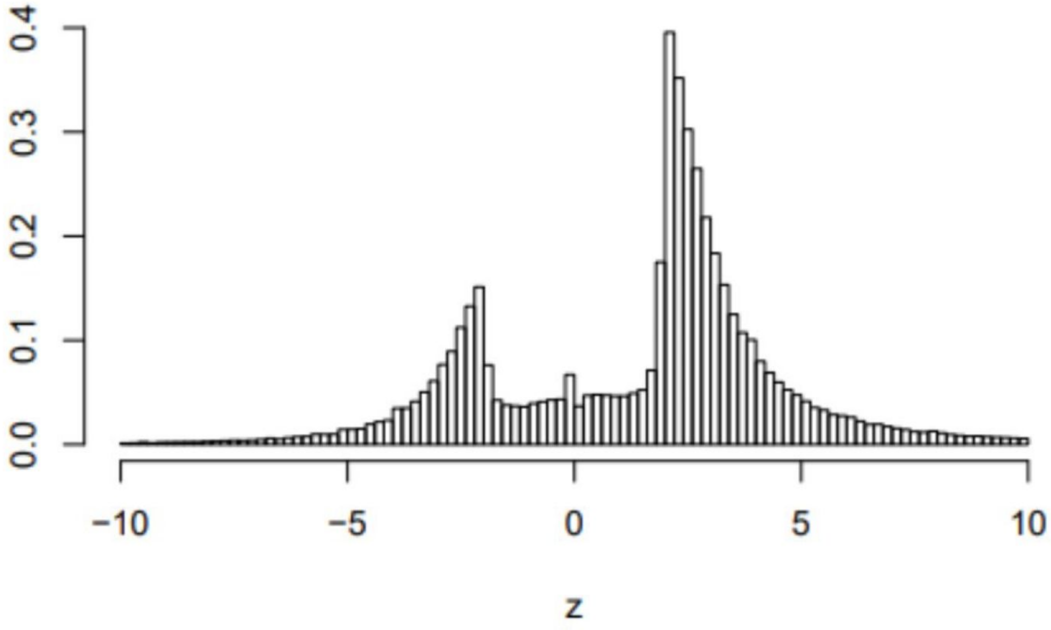
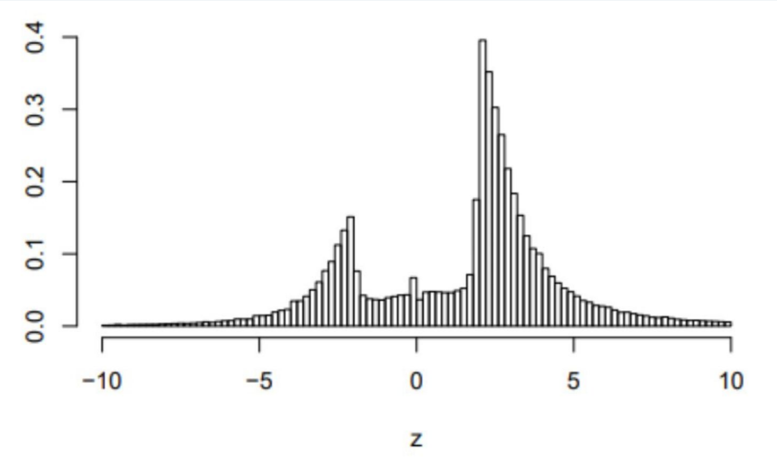
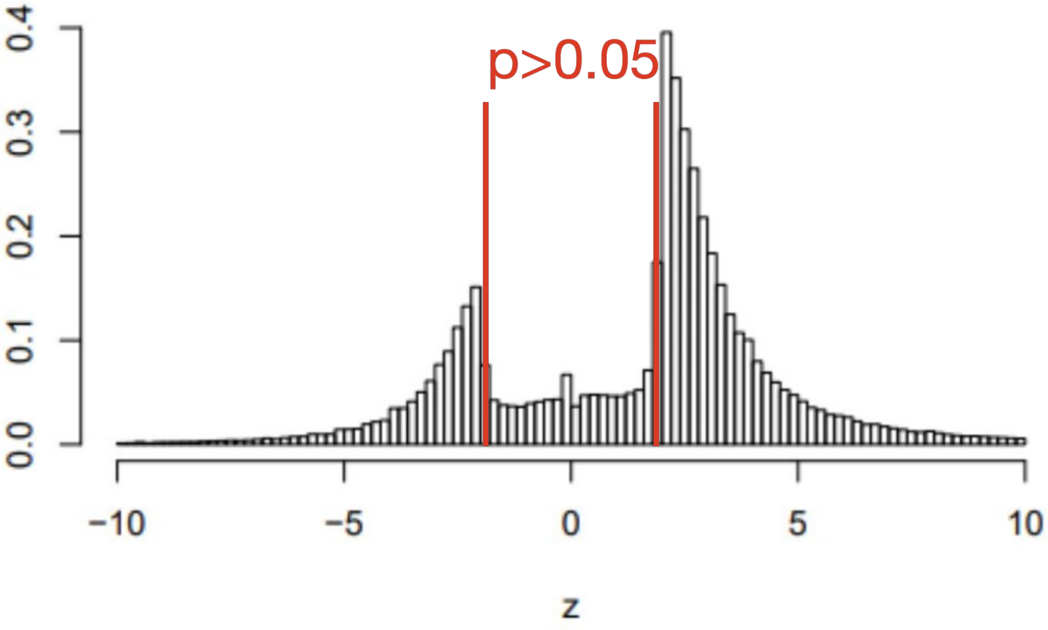


Figure 1: The distribution of more than one million z-values from Medline (1976–2019).

Think-pair-share: p-values in published research



What explains this distribution?



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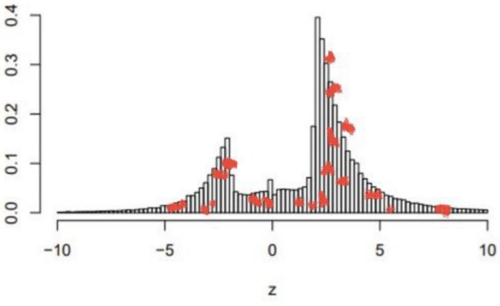

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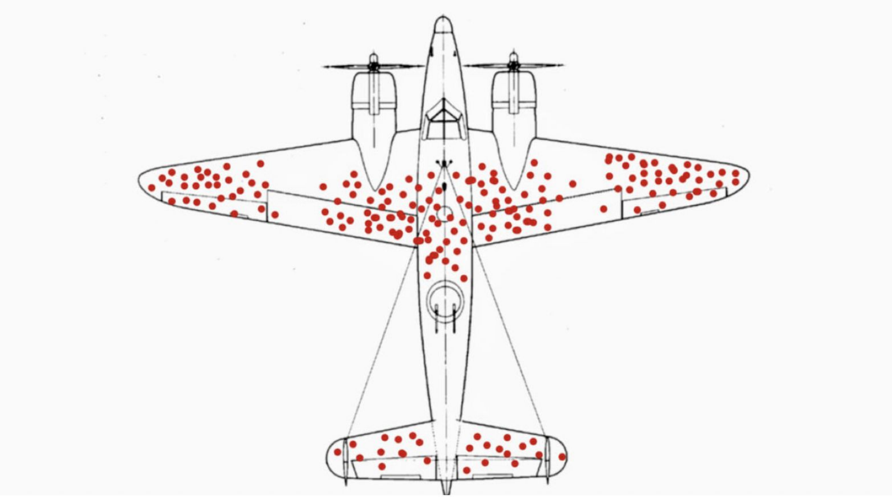

How can we be more right and less wrong?

- Be aware of our biases; use external review to check and minimize
- Think about tools; know their strengths and limitations
- Analysis and research require rigorous methods (and time)
- Social factors influence decision making; communication is part of the process
- Stay curious; question yourself and your team throughout